

Generalizing Human Motion Style Transfer Method Based on Metadata-independent Learning

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Abstract

This study aims to extend the applicability of motion style transfer methods to be robust for diverse and complex motions akin to those found in real-world data. We introduce a new training method without style labels which are metadata needed by conventional motion style transfer methods and enable a model to learn from almost motion datasets. This contribution mitigates performance degradation for motions absent in style-motion datasets. Compared to conventional methods, the proposed method demonstrates superior transfer performance, particularly under conditions featuring more diverse and complex motions.

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1 Introduction

3D human motion editing methods enable us to make 3D avatar more vivid and transforms the 3D animation into something realistic and lifelike without many costs. One of them, motion style transfer methods can modify human motions with desired styles expressing characters and emotions and many methods have been proposed [Aberman et al. 2020; Tao et al. 2022].

However, these conventional methods have difficulties to be generalized because of the datasets' limitation. Specifically, the style-motion datasets for the style transfer task consist of only simple motions because it needs high cost to obtain 3D human motions such as measurement by motion capture and estimation from multi-view videos. In addition, conventional style transfer

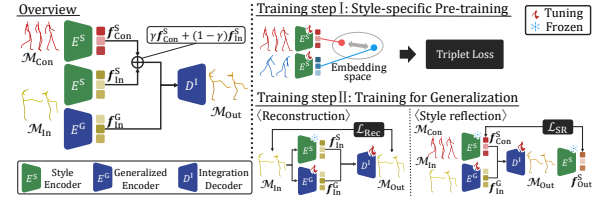


Figure 1: Overview of the proposed method.

methods need metadata, style labels, for all motions in their training step and cannot utilize other tasks' motion datasets correctly. Consequently, the representation which the conventional methods can learn is inherently limited and these methods have limitations in transferring diverse and complex motions with high performance.

In this paper, to realize the generalized motion style transfer method, we focus on metadata-independent learning. Specifically, we design the model around a style-specific network capable of extracting style attributes clearly, alongside generalized networks that consider fundamental attributes and generate motions reflecting both attributes. We train the generalized networks from various motion datasets with or without style labels, in addition to training a task-specific network following conventional approaches. Consequently, the proposed method can mitigate performance degradation for motions that are not contained in the conventional style-motion datasets.

2 Proposed Method

The proposed generalized motion style transfer method aims to achieve high performance for diverse and complex motions that are absent in style-motion datasets. An overview of the proposed method is illustrated in Fig. 1. The proposed model consists of a style encoder E^S for extracting style features, a generalized encoder E^G for extracting general features, and an integration decoder D^I for generating motion data reflecting both sets of features. Note that, the general features express fundamental attributes associated with all motion tasks, such as actions. In the proposed method, we extract the style features of the input motion M_{In} and the conditional

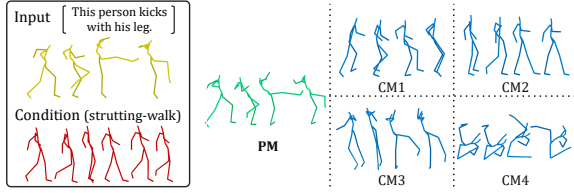


Figure 2: Generation examples of each method. PM can reflect the style of the conditional motion as shown that generated motions kick with one’s chest held high.

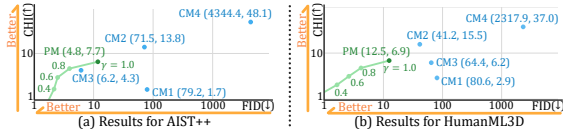


Figure 3: FID and CHI of each method.

motion $\mathcal{M}_{\text{Con}}$ and add them using the weight γ , which represents the degree of style reflection. Then we input calculated style feature and the general feature of $\mathcal{M}_{\text{In}}$ to D^I and obtain transferred motion $\mathcal{M}_{\text{Out}}$. We train the proposed model in two steps: style-specific pre-training and training for generalization.

In the first step, we pre-train E^S using the only style-motion datasets. Specifically, we calculate the triplet loss related to style labels and pre-train E^S . Since the style-motion datasets provide clear style attributes, E^S learns to extract features containing rich information pertinent to the motion style transfer.

In the second step, we train E^G and D^I by focusing on style preservation and attribute reflection. Specifically, we train the model based on the reconstruction and transfer tasks. Because the model is expected to reconstruct the input motion when the same motion is used as both input and condition, we calculate the reconstruction loss $\mathcal{L}_{\text{Rec}} = \|\mathcal{M}_{\text{In}} - \mathcal{M}_{\text{Out}}\|_2$. This concept resembles that of an AutoEncoder, contributing to enhancing the robustness of the proposed method for motion with or without specific metadata.

Moreover, the generated motion should incorporate the same style attributes as the conditional motion. We compute the style-reflection loss $\mathcal{L}_{\text{SR}} = \|f_{\text{Con}}^S - f_{\text{Out}}^S\|_2$ based on the style-specific features of the conditional motion f_{Con}^S and the transferred motion f_{Out}^S . Using the pre-trained E^S for feature extraction and style preservability evaluation, the proposed method can learn style reflection without relying on style labels.

Finally, we train E^G and D^I based on the final loss $\mathcal{L}_{\text{Final}} = \mathcal{L}_{\text{Rec}} + \mathcal{L}_{\text{SR}}$. By introducing reconstruction and transfer tasks without style labels, we achieve unsupervised training for networks. Consequently, the proposed method can learn from almost motions, with or without style labels, and its robustness is improved.

3 Experiments

To validate the proposed method’s generalized ability, we conducted experiments focusing on the style transfer task using more diverse datasets than those employed in existing studies.

To validate the style transfer performance on diverse and complex motions not represented in motion-style datasets, we augmented the popular motion-style dataset introduced by Xia *et al.* [Xia et al. 2015] with motions from HumanML3D [Guo et al. 2022] and AIST++ [Li et al. 2021], respectively. The dataset by Xia *et al.* assigns one label from eight styles to each motion, HumanML3D comprises numerous motions, and AIST++ comprises complex dance motions.

As the comparative methods (CMs), we selected state-of-the-art motion style transfer methods that could be adapted to our task such as Aberman’20 [Aberman et al. 2020] and style-ERD [Tao et al. 2022]. Because conventional methods cannot learn from the motions without style labels, we designated the methods trained solely on labeled motions from the dataset provided by Xia *et al.* as CM1 (Aberman’20) and CM2 (style-ERD). Furthermore, we artificially assigned “neutral” label to unlabeled motions in HumanML3D and AIST++ and trained models using all motions as CM3 (Aberman’20) and CM4 (style-ERD) to ensure fair evaluation conditions.

We evaluated proposed method (PM) and CMs based on two perspectives: action preservation and style reflection. To evaluate action preservation, we conducted quantitative assessments using the Fréchet inception distance (FID) metric, which is commonly used in previous motion-style transfer tasks. Additionally, we assessed style reflection using the Calinski-Harabasz index (CHI) that is typically employed in clustering tasks because it is desirable for style transfer methods to ensure that style-focused features of each of the eight styles form distinct clusters.

We show the qualitative results in Fig. 2, the quantitative results in Fig. 3. From Figs. 2 and 3, we confirm that PM can achieve action preservation and style reflection better than CMs. Note that, although CM2 and 4 exhibit higher CHI than PM, it is considerably inferior for the action preservation as shown in Fig. 2, making it unsuitable for these experiments’ task. Therefore, we confirm the effectiveness of the proposed method for action preservation and style reflection in the style transfer tasks. Furthermore, Fig. 3 show that indicates that the proposed method assigns style to the output while maintaining a low FID by increasing γ .

4 Conclusion

In this paper, we extend the applicability of motion style transfer methods to be robust for diverse and complex motions. By introducing the training method that do not depend on style labels, the proposed method can learn from almost motions in multiple datasets and its robustness can be increased. The experimental results demonstrate that the proposed method achieves greater robustness for diverse and complex motions than conventional methods.

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